

Contextual Embeddings for Financial Sentiment Analysis

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1 Introduction

This report details the implementation and evaluation of a sentiment analysis pipeline for financial news, using the pretrained `prajjwal1/bert-mini` transformer encoder. The study compares various embedding extraction strategies and downstream classification architectures against established baselines like VADER and CBOW.

2 Experimental Results

2.1 F1 Score Comparison

Table 1: Macro F1-score comparison across all models.

Setup	Macro F1-score
VADER (Baseline)	0.4931
CBOW + LogReg (Reference)	0.68 – 0.74
Setup 1: Frozen CLS + LogReg	0.7325
Setup 2: Frozen Mean Pooling + LogReg	0.7238
Setup 3: Frozen Mean Pooling + MLP	0.7943
Setup 4: End-to-End Fine-Tuning	0.9312

2.2 Confusion Matrix (Best Model)

Table 2: Confusion Matrix for Fine-Tuned BERT-mini.

True \ Pred	Negative	Neutral	Positive
Negative	59	0	2
Neutral	1	271	6
Positive	9	4	101

3 Contextual Representation Analysis

3.1 "Bank" Disambiguation Probe

To assess the encoder’s ability to capture semantics, we compared the embeddings of the word "bank" in two distinct contexts:

1. *“The bank raised interest rates.”*
2. *“She sat by the river bank.”*

The cosine similarity results are as follows:

- **CLS Pooling:** 0.7158
- **MEAN Pooling:** 0.6780

Both strategies distinguish the two senses, as the similarity is significantly below 1.0. Mean pooling appears slightly more sensitive to the local context changes, yielding a lower similarity score. In contrast, a static CBOW embedding would produce a similarity of 1.0 for the word “bank” regardless of context, as it assigns a single vector to each word type. This demonstrates the superior representational power of Transformer-based contextual embeddings.

4 Error Analysis

4.1 BERT Success / VADER Failure

Contextual embeddings excel in identifying domain-specific nuances that lexicon-based tools like VADER miss. BERT’s ability to handle complex negations and multi-word financial expressions gives it a distinct advantage over VADER’s rule-based approach.

4.2 BERT Failure / VADER Success

BERT-mini, being a smaller model, can occasionally fail on extremely short, sarcastic, or highly literal phrases where a simple lexicon match is more robust. For instance, a very short sentence like *“Profit is flat.”* might be misinterpreted as neutral or slightly positive by BERT whereas VADER might correctly assign it a neutral score based on its predefined dictionary.

5 Fine-Tuning vs. Frozen Embeddings

Fine-tuning the entire encoder provides a massive F1-score improvement (0.9312 vs. 0.7325 for frozen CLS). However, this comes with increased compute cost. Setup 3 (Mean + MLP) offers a balance, outperforming simple linear classifiers by capturing non-linear relationships in the frozen feature space. For production systems with strict latency requirements, using frozen embeddings with a fast linear head might be preferable, but for maximum accuracy in financial sentiment, fine-tuning is the best.